

Islamic University of Technology

Course Code : EEE 4308

Course Name : Digital Electronics Lab

Project Name : SAP-1 Architecture-based 8-bit Computer Design and Implementation

Phase : B

Team : Byte_Masters

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Project Objective:

Objective: The overall goal of this project is to design and implement an 8-bit computer based on the SAP-1 (Simple-As-Possible) architecture using a combination of combinational and sequential digital circuits.

Brief Description of Project:

The system follows a single 8-bit bus architecture for efficient data transfer and includes essential components such as RAM, registers, ALU, control unit, program counter, and input/output units to execute basic instructions.

In the previous phase (Phase A) of the project, the main clock system with both Auto and Manual modes of operation was successfully implemented along with the Program Counter, establishing the timing and address sequencing foundation of the computer.

In this phase, the focus shifts to the Memory and Register section of the system. The work in this report concentrates specifically on the implementation and explanation of the submodules:

- RAM (16X8)
- Registers: Accumulator (Register A) and Register B

These submodules are essential for storing data, transferring data through the bus, and supplying operands to the ALU during instruction execution. This phase aims to design, analyze, and demonstrate the working of these components in accordance with the SAP-1 architecture and the constructed circuit.

Submodules:

Registers

Accumulator

Accumulator (Register A)

The Accumulator, also known as Register A, is one of the main registers in the SAP-1 computer. It is used to store data that is involved in arithmetic and logic operations performed by the ALU.

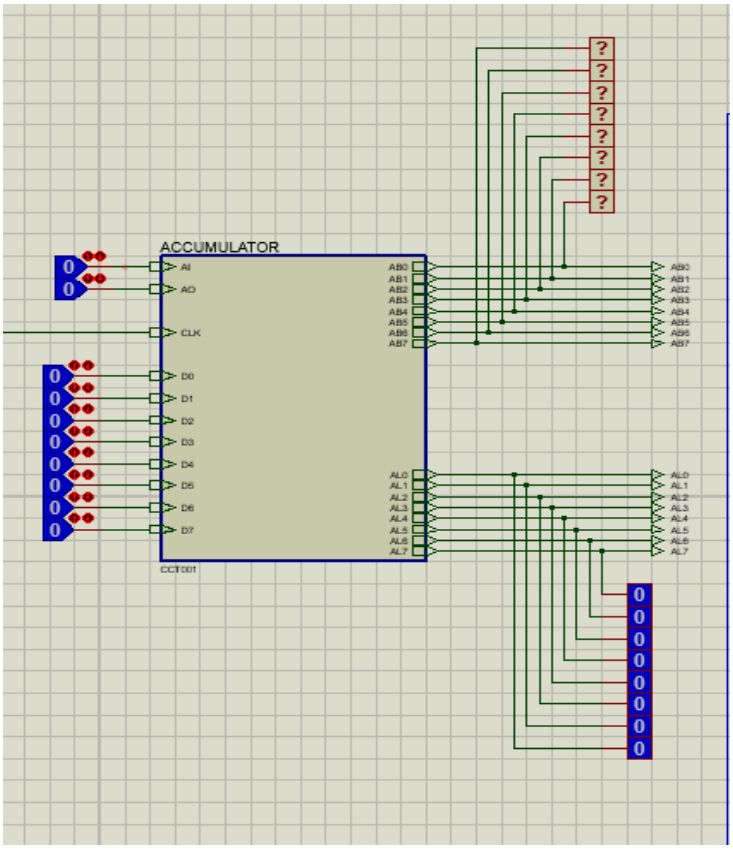


Fig: Block diagram of Accumulator

During instruction execution, the Accumulator holds one operand for the ALU and also stores the result after the operation.

In this system, the Accumulator is an 8-bit register that can load data from the common bus, hold it, and place it back onto the bus when needed. This operation is controlled using the AI and AO signals, which ensure proper data transfer without conflict on the bus.

Because it stores data from memory and also receives results from the ALU, the Accumulator plays a central role in the data processing of the SAP-1 computer.

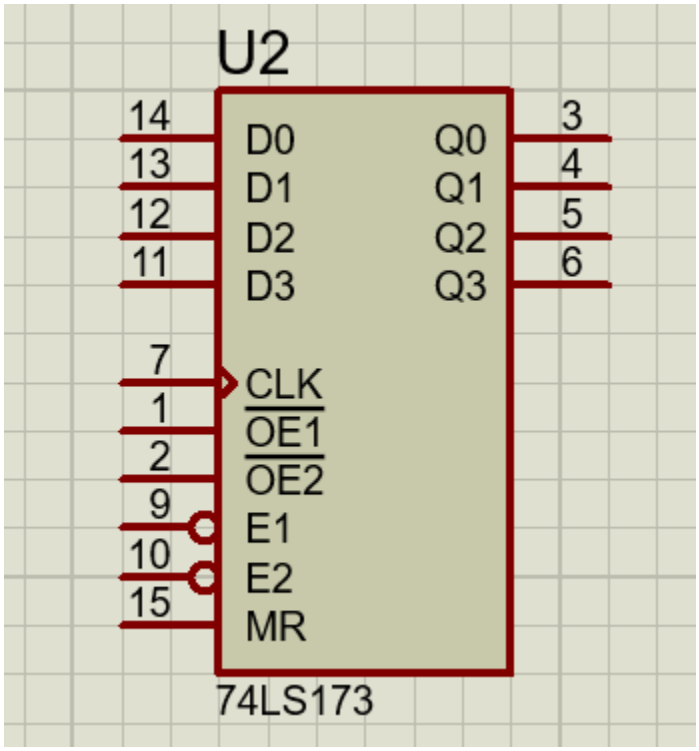
IC Components Used in Accumulator Circuit

Based on the designed circuit of the Accumulator (Register A), the following integrated circuits are used to implement the 8-bit register, control the data flow, and interface it with the common bus of the SAP-1 computer.

Integrated Circuits (ICs):

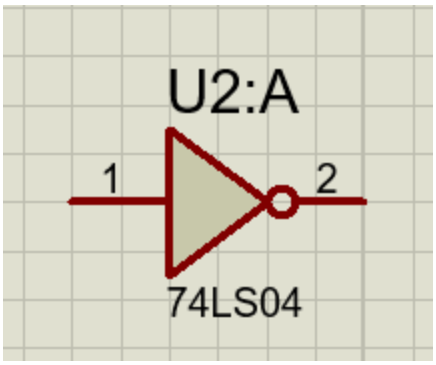
○ **74LS173 – 4-bit D Register with 4-State Output (U18 and U19)**

Two 74LS173 ICs are used to construct the 8-bit Accumulator.



- Each 74LS173 stores 4 bits of data using D flip-flops.
- They are positive-edge-triggered.
- Together, U18 and U19 form the complete 8-bit register**.**
- The D inputs (D0–D7) are connected to the common bus to load data.
- The CLK pin allows data to be stored on the clock edge when the Load signal is active.
- The MR (Master Reset) pin is used to clear the Accumulator when required.
- The OE (Output Enable) pins control when the stored data is allowed to leave the register.
- E1 and E2 are the enable pins. They are active low.

These ICs are responsible for storing the data that comes from the bus and holding it until the next operation.

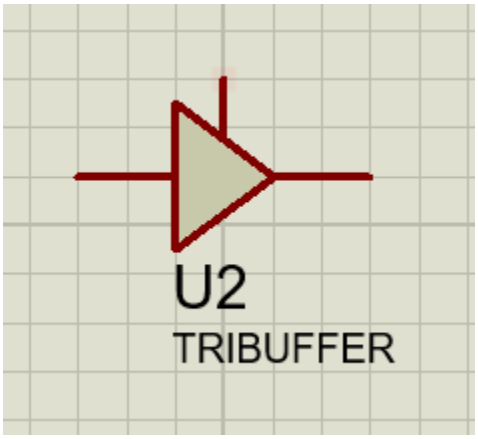


○ **74LS04 – Hex Inverter (U28:A)**

A 7404 inverter is used to invert the control signal/load signal (AI) for proper activation of the register.

- It ensures the correct logic level reaches the Enable/Load pins of the 74LS173.
- This is necessary because some control signals in SAP-1 are active low**,** while the register requires the opposite logic.

This IC helps in matching the control signal logic with the register's requirement.



Tri-State Buffers (U20–U27)

Eight tri-state buffers are used at the outputs of the registers before connecting to the bus lines (AB0–AB7).

- These buffers allow the Accumulator to place data onto the bus only when Output Enable is active**.**
- When disabled, they remain in a high-impedance state, preventing bus conflict with other modules.
- Each buffer corresponds to one bit of the 8-bit output.
- These buffers ensure safe and controlled data transfer from the Accumulator to the common bus.

Logic Connections and Control Lines:

- **CLK Line:** Connected to both 74LS173 ICs to synchronize data storage.
- **Load Signal:** Allows data from the bus to be stored into the Accumulator.
- **Clear (MR):** Resets the stored value to 0 when required.
- **Output Enable:** Controls when the Accumulator sends data to the bus.

Together, these components allow the Accumulator to load data from the bus, store it reliably, and send it back to the bus when required during ALU operations.

Description of Connection:

In the design, the Accumulator is implemented in a child sheet and connected to the rest of the SAP-1 system through the parent sheet.

Connections of the Parent Sheet

Input Pins

D0–D7:
These eight lines come directly from the common bus. The 8-bit value present on the bus is fed into the Accumulator through these pins.

CLK (Clock):
The clock input of the Accumulator is connected to the same clock signal used in the parent sheet. This common clock connection ensures that the Accumulator operates in synchronization with the rest of the implemented circuit. On the active clock edge, the Accumulator stores the data present on D0–D7 when AI is active.

AI (Accumulator In):
This control signal decides when the Accumulator should accept data. When AI is active, the data coming from RAM through D0–D7 is stored into the Accumulator on the next clock pulse.

AO (Accumulator Out):
This signal controls when the Accumulator is allowed to send its stored data to the next stage of the circuit. When AO is active, the stored value is available at the output lines and it transfer the values to the bus.

Output Pins

AB0–AB7:
These lines carry the 8-bit stored value from the Accumulator towards the bus section of the circuit. It operates only when AO is high.

AL0–AL7:
These lines carry the outputs of bus towards ALU through Accumulator.

Connections inside the Child Sheet

Data Inputs to Registers (U18 and U19 – 74LS173)

The 8-bit input from D0–D7 is divided into two groups:

- D0–D3 → U18
- D4–D7 → U19

Each 74LS173 stores 4 bits, together forming the 8-bit Accumulator.

Clock (CLK)

The CLK line is connected to both U18 and U19 so both ICs store data at the same time, behaving as a single 8-bit register.

AI Control with 7404 Inverter

The AI signal first passes through the 7404 inverter (U28:A)**.**
The inverted signal is then connected to the enable pins of both 74LS173 ICs.

This is required because the 74LS173 uses active-low enable, while AI is active-high.

Register Outputs (AL0–AL7)

The outputs Q0–Q7 from U18 and U19 are labeled as AL0–AL7**.**
These lines carry the stored data from the registers to the tri-state buffers.

Tri-State Buffers to Outputs (U20–U27)

The outputs from the two 74LS173 registers are labeled as AL0–AL7**.** These lines do not go directly to the final outputs. Instead, they first pass through a set of tri-state buffers (U20–U27)**.**

The purpose of these buffers is to control when the Accumulator is allowed to send its stored data forward in the circuit. The outputs of these buffers form the lines AB0–AB7**,** which carry the 8-bit value of the Accumulator to the next stage.

AO Control to Buffers

All the tri-state buffers are controlled by the AO signal.

- When AO is active, the stored Accumulator value passes to AB0–AB7.
- When AO is inactive, the buffers remain in high-impedance state.

Through these connections, the Accumulator receives data from RAM**,** stores it using clock and AI control, and forwards the stored value to the ALU side through AO when required.

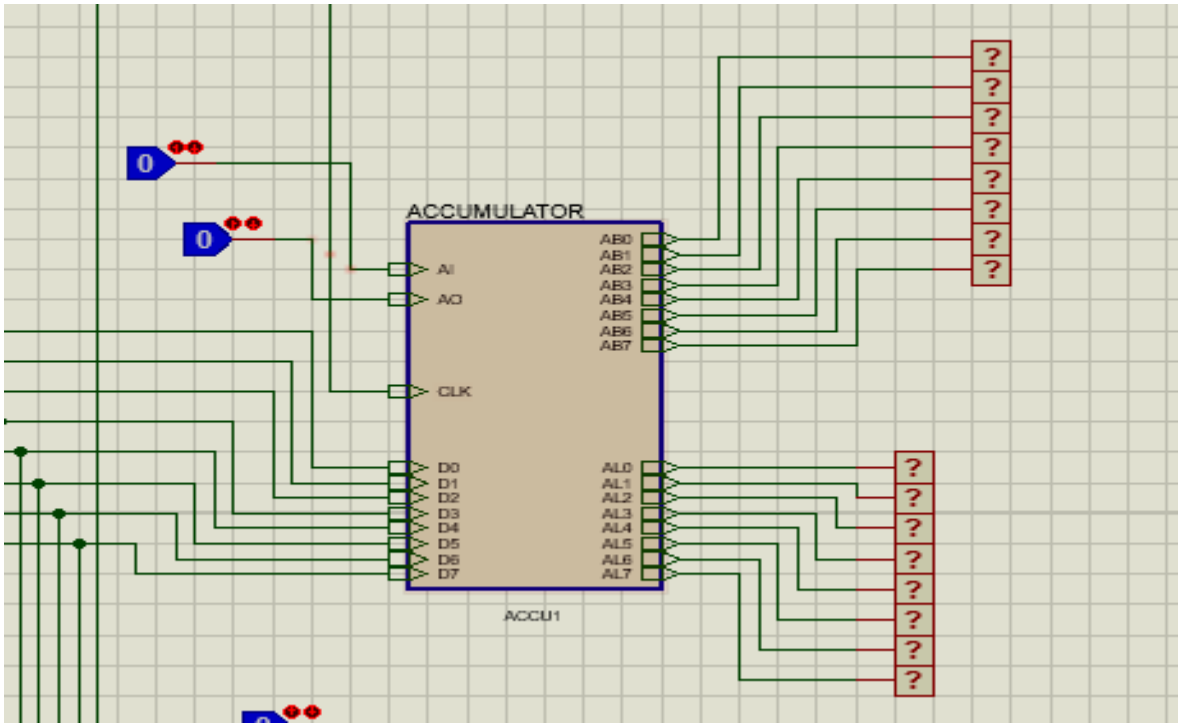


Fig: Parent Sheet of Accumulator

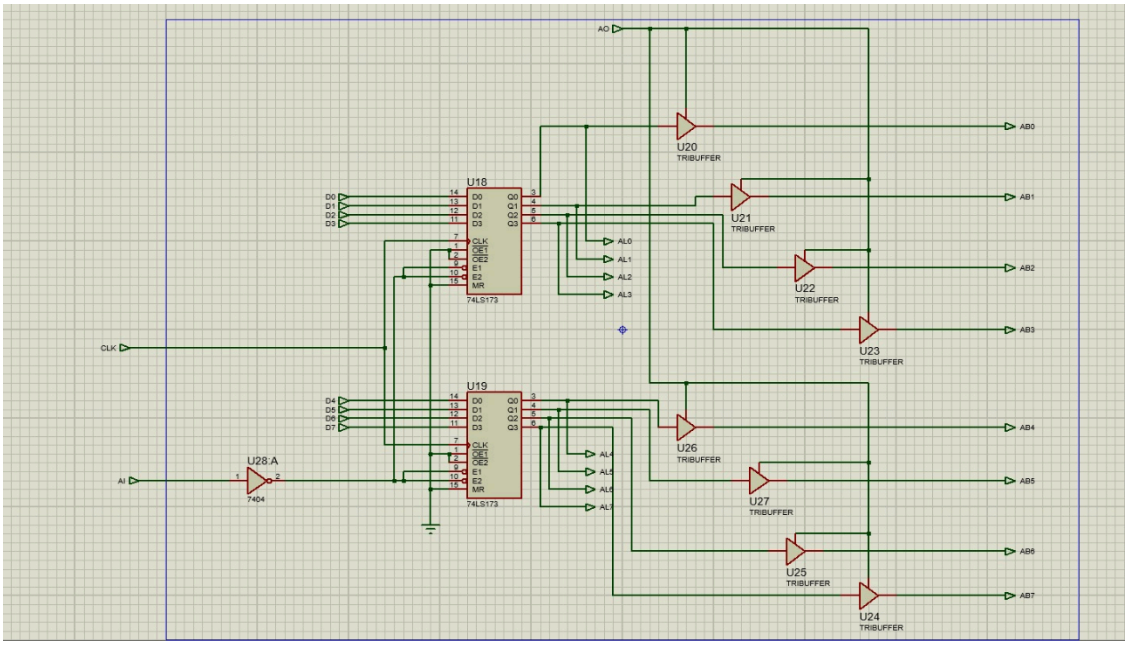


Fig: Child Sheet of Accumulator

Functions

The Accumulator is an essential component of the SAP-1 architecture that serves as the primary data-holding register for arithmetic and logical operations performed by the system.

It temporarily stores data retrieved from memory or transferred through the system bus, allowing the processor to manipulate and reuse values during instruction execution. By maintaining intermediate results, the Accumulator enables multi-step computations to be performed in an organized and sequential manner.

The Accumulator holds its contents as an 8-bit binary value, which can be used directly by the Arithmetic Logic Unit (ALU) as one of its main operands. This allows operations such as addition or subtraction to be carried out efficiently, with the results typically written back into the Accumulator for further processing or output.

Additionally, the Accumulator can place its stored value onto the system bus when required, making the data available to other modules such as memory, registers, or output devices. This controlled data sharing ensures coordinated communication between different parts of the architecture.

Overall, the Accumulator plays a central role in SAP-1 operation by storing working data, supporting computation, and facilitating data transfer across the system, thereby contributing to efficient and structured instruction execution.

Working Principle

When the Clock (CLK) signal pulses, the Accumulator will load data from the system bus into its internal register if both enable inputs (E1 and E2) are active. These enable signals allow the incoming bits on the data lines to pass into the flip-flops of the register. So, let's say the Accumulator initially stores `00000000`. When valid data appears at the inputs and a Clock (CLK) pulse arrives while E1 and E2 are enabled, each flip-flop updates its state and captures the new bits, storing them as the current value of the Accumulator.

However, because of small propagation delays inside the flip-flops and logic gates (in the nanosecond range), the change does not occur instantly; the signals require a short time to stabilize before the final stored value appears at the outputs.

The stored value is continuously available at the register's output pins, since the internal output enables are tied active. But the value is only placed onto the shared system bus when the Accumulator Output control signal (AO) activates the tri-state buffers. When AO is active, the buffers allow the stored data to drive the bus lines. When AO is inactive, the buffers enter a high-impedance state, effectively isolating the Accumulator so that other modules may use the bus without conflict.

If a reset condition is applied through the master reset line, the register clears its contents back to `00000000`, removing previously stored data. The Accumulator then continues operating in synchronization with the clock, repeatedly capturing new input values when enabled and

supplying its stored data to the system when commanded, supporting arithmetic and data transfer operations within the SAP-1 architecture.

Register B

Register B is an important storage element within the SAP-1 architecture that temporarily holds operand data used by the Arithmetic Logic Unit (ALU). It works alongside the Accumulator by providing the second input required for arithmetic operations. The register captures data from the system bus when enabled and supplies its stored value directly to the ALU lines, ensuring synchronized and organized data flow during instruction execution.

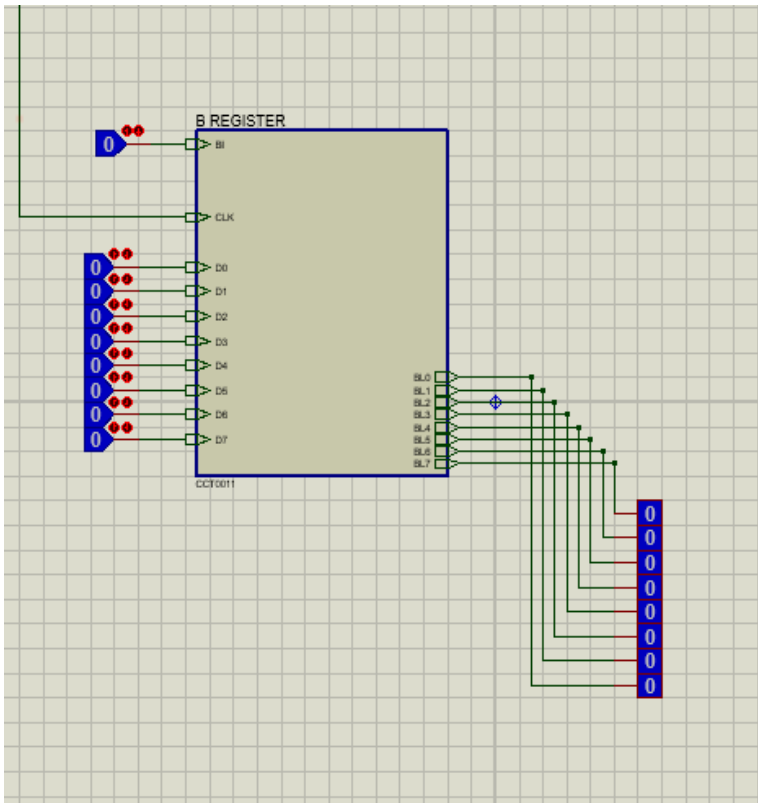


Fig: Block diagram of Register B

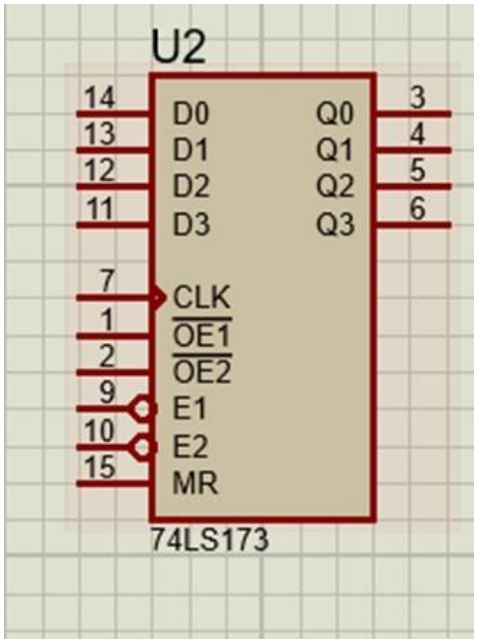
IC Components Used in Register B Circuit

Based on the designed circuit of Register B**,** the following integrated circuits are used to implement the 8-bit storage register,

Integrated Circuits (ICs):

○ 74LS173 – 4-bit D Register with 4-State Output (U29 and U30)

Two 74LS173 ICs are used to construct the 8-bit Register B.

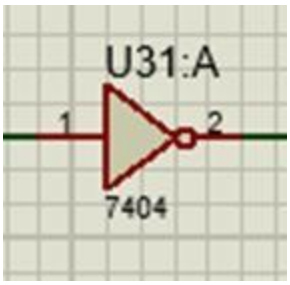


- Each 74LS173 stores 4 bits of data using edge-triggered D flip-flops.
- They operate on the positive edge of the clock signal.
- Together, U29 and U30 form a complete 8-bit storage register.
- The D inputs (D0–D7) receive operand data from the system bus.
- The CLK pin allows data to be stored on the clock edge when the load signal is active.
- The MR (Master Reset) line clears stored data when required.
- The OE pins control the availability of register outputs internally.
- Their outputs are routed directly toward the ALU input lines.

These ICs are responsible for capturing operand data from the bus and holding it steady for arithmetic operations.

○ 74LS04 – Hex Inverter (U31:A)

A 7404 inverter is used to invert the control/load signal before it reaches the enable pins of the registers.



- It adjusts the logic level of the Register B input control signal.
- Ensures compatibility with the active-low enable behavior of the 74LS173.
- Guarantees that loading occurs only when the intended control condition is met.

This IC ensures correct signal polarity between the control unit and the register hardware.

Logic Connections and Control Lines:

- **CLK Line:** Connected to both 74LS173 ICs to synchronize storage of data.
- **Load Signal:** Determines when Register B captures operand data from the bus.
- **Clear (MR):** Provides resetting of stored contents when necessary.
- **ALU Output Lines:** Carry stored register contents to the ALU for processing.

Together, these components allow Register B to receive operand data, store it reliably, and provide it directly to the ALU when required for computation.

Description of Connection

In the design, Register B is implemented in a child sheet and connected to the SAP-1 system through the parent sheet, allowing it to receive data from the bus and supply operands to the ALU

Connections of the Parent Sheet

Input Pins

D0–D7:
These eight lines originate from the system bus and provide operand data to Register B. The values transferred through these lines are stored when the load control signal is active.

CLK (Clock):
Connected to the common system clock used throughout the SAP-1 circuit. This ensures Register B stores incoming data in synchronization with all other modules.

Control Signal (Register B In):
This signal determines when Register B accepts data. When active, the value present on D0–D7 is captured on the next clock edge.

Output Pins

AL0–AL7:
These lines carry the stored 8-bit value directly to the ALU input interface. The outputs serve as the second operand for arithmetic operations alongside the Accumulator data.

Connections Inside the Child Sheet

Data Inputs to Registers (U29 and U30 – 74LS173)

The incoming 8-bit data is divided into two sections:

- D0–D3 → U29
- D4–D7 → U30

Each IC stores four bits, and together they behave as a single 8-bit register.

Clock Distribution

The CLK signal is connected to both U29 and U30 so data storage occurs simultaneously across all bits, ensuring consistent operand representation.

Control Signal Through Inverter(BI)

The load/control signal passes through inverter U31:A before reaching the enable pins of both registers.

- This inversion flips the signal so it has the correct polarity. Since the chips are "active-low," they need a "0" to turn on, while the system sends a "1" to load. The inverter makes these two signals work together.
- Allows the registers to load data only when commanded by the control logic

Register Outputs (AL0–AL7)

Outputs Q0–Q7 of U29 and U30 are labeled AL0–AL7.

- These lines carry stored data toward the ALU
- No tri-state buffering is used here since Register B feeds the ALU directly rather than the shared system bus

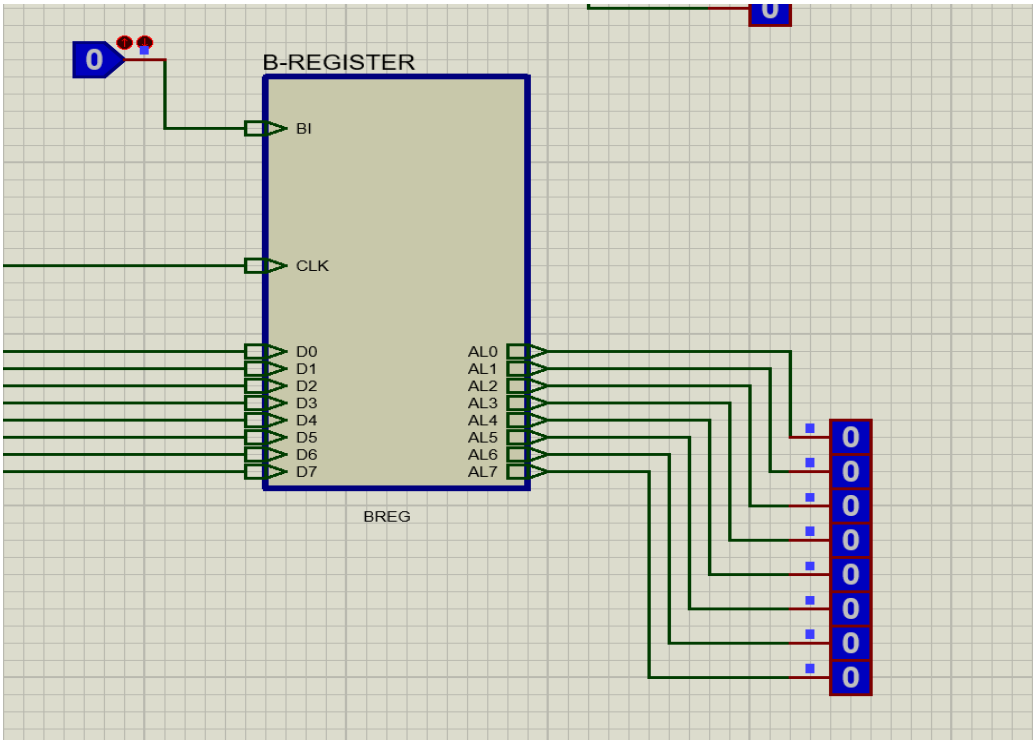


Fig: Parent Sheet of Register B

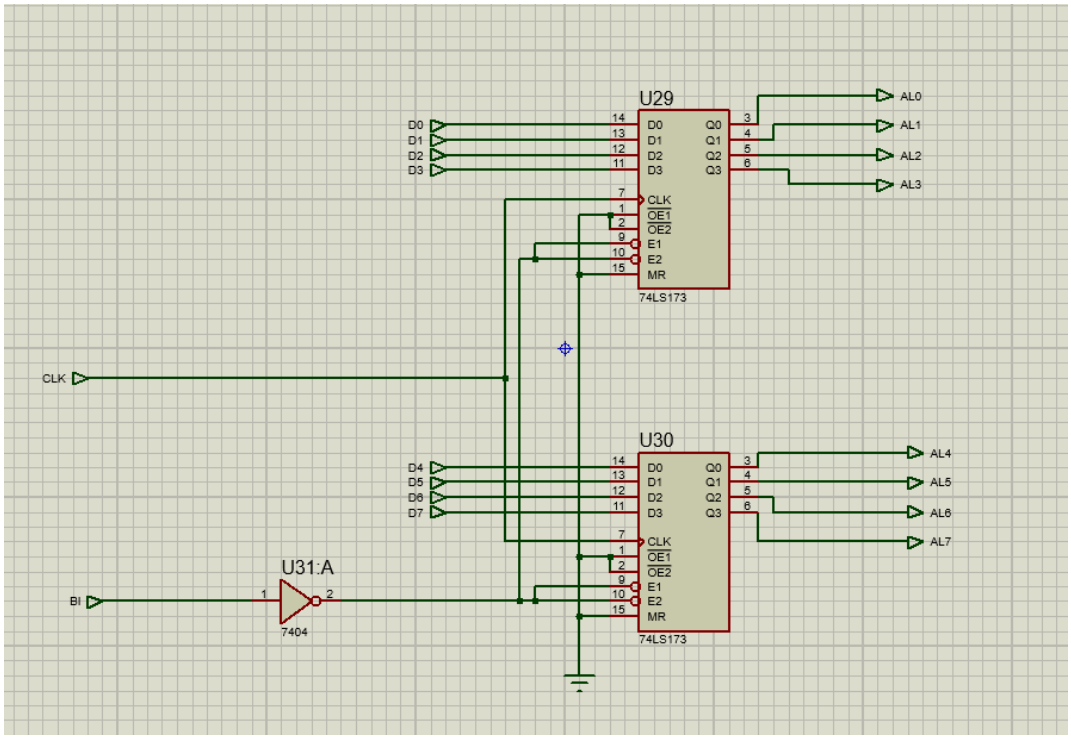


Fig: Child Sheet of Register B

Function

Register B is an essential storage element within the SAP-1 architecture that provides the second number required by the Arithmetic Logic Unit (ALU) for arithmetic operations.

It temporarily stores data transferred from the system bus and supplies this data directly to the ALU when instructed by the control unit. By holding the second value separately from the Accumulator (which holds the first number), Register B enables the execution of instructions such as addition and subtraction where two data inputs are required.

The register stores its contents as an 8-bit binary value, ensuring compatibility with the data width of the SAP-1 system. This stored value remains stable until a new value is loaded, meaning the processor can reuse that same number for several steps if it needs to.

Unlike the Accumulator, Register B does not place its data back onto the common system bus. Instead, its output is dedicated to the ALU input lines, ensuring that the data is always available for computation without causing contention on shared communication paths.

Basically, Register B acts as a reliable "holding area" for numbers. It allows the ALU to combine data from the Accumulator and memory, thereby contributing to the computational capability of the SAP-1 system.

Working Principle

When the Clock (CLK) signal pulses, Register B captures data from the input lines if its load control signal is active. The register is implemented using two 74LS173 4-bit D registers (U29 and U30), which together form an 8-bit storage unit. For example, if a binary value is present on lines D0–D7 and the load signal is active, the next clock pulse causes each internal flip-flop to store its corresponding bit, updating the contents of the register simultaneously.

Because of the propagation delays inside the flip-flops and logic gates, the state change is not instantaneous; the stored value stabilizes shortly after the clock edge as signals propagate through the devices.

The load signal first passes through an inverter before reaching the enable inputs of both registers. This inversion ensures the signal to be compatible with the active-low enable behavior of the 74LS173 (meaning they need to be 0 to enable), allowing data to be stored only under the intended control condition. If the load signal is inactive, the enable inputs prevent new data from entering, and the register holds onto whatever data they already had.

The clock line is shared between both 74LS173 ICs so that all eight bits are updated in synchronization, allowing them to behave as a single register. The master reset connection provides the ability to clear stored contents when required, forcing all output bits to zero.

Once data is stored, the outputs from U29 and U30 appear on lines AL0–AL7. These lines are connected directly to the ALU. Since Register B does not use tri-state buffers for bus access, its

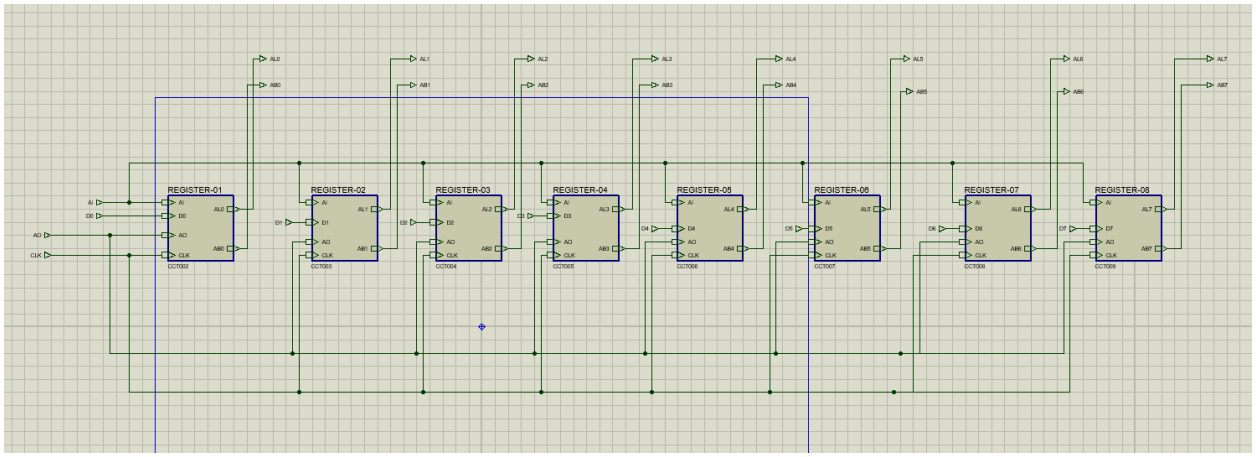
outputs remain dedicated to the ALU. This ensures the ALU always has a steady, reliable number to work with during calculations.

Through this operation, Register B reliably captures data on clock events, preserves it until updated or cleared, and supplies it as a stable operand to the ALU during instruction execution.

The Accumulator and Register B can be built using basic logic gates instead of using any ICs. Each one is made from 8 identical 1-bit register cells to form an 8-bit register. Both registers connect to the common bus and work with control signals and a shared clock. The Accumulator stores data and can send it back to the bus when needed, while Register B holds data for later use in the system.

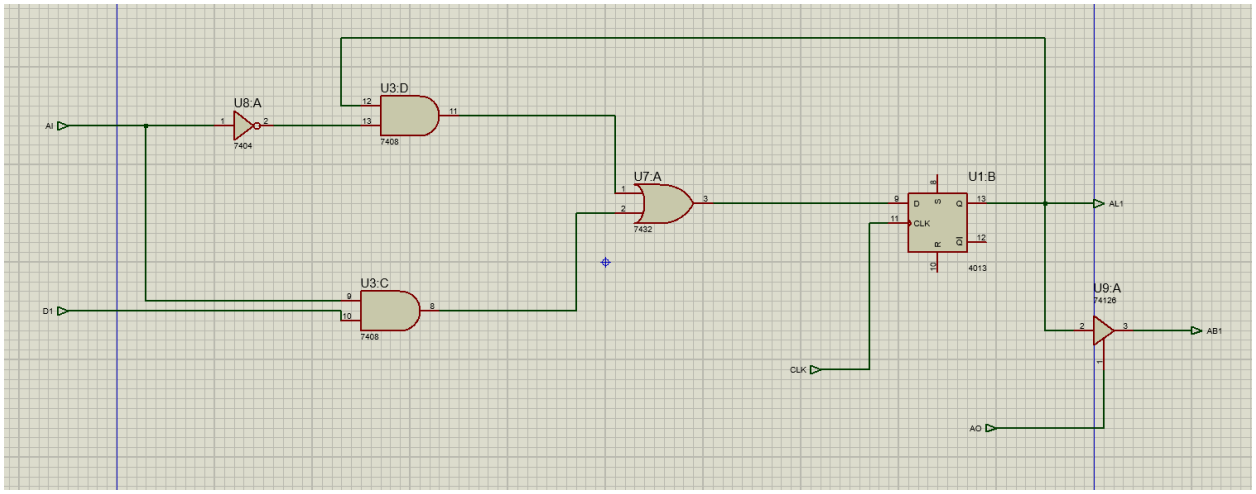
Accumulator (Implemented using Logic Gates)

Connection in Parent Sheet



In the parent sheet, the Accumulator is shown as an 8-bit register connected to the common bus. The lines D0–D7 bring data from the bus into the Accumulator. The control signal AI decides when the Accumulator should load this data, and AO decides when the stored value should be placed back onto the bus through AB0–AB7. The CLK line is common to all bits and ensures that data is stored only on the clock edge.

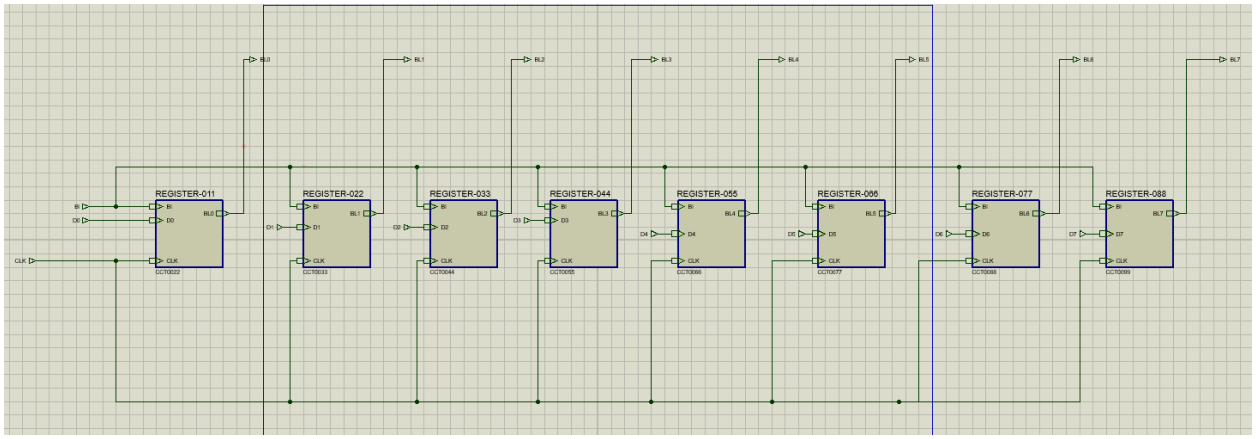
Connection in Child Sheet



Inside, the Accumulator is built using eight identical 1-bit register cells made from logic gates, a 4013 D flip-flop**,** and a 74126 tri-state buffer. The data input and the AI signal pass through logic gates (7404, 7408, 7432) to control whether the data is allowed to reach the flip-flop. On the clock edge, the flip-flop stores the bit. The stored output appears on the AL line. When AO is active, the tri-state buffer allows this stored bit to go to the bus line (AB)**.** Repeating this cell eight times forms the 8-bit Accumulator.

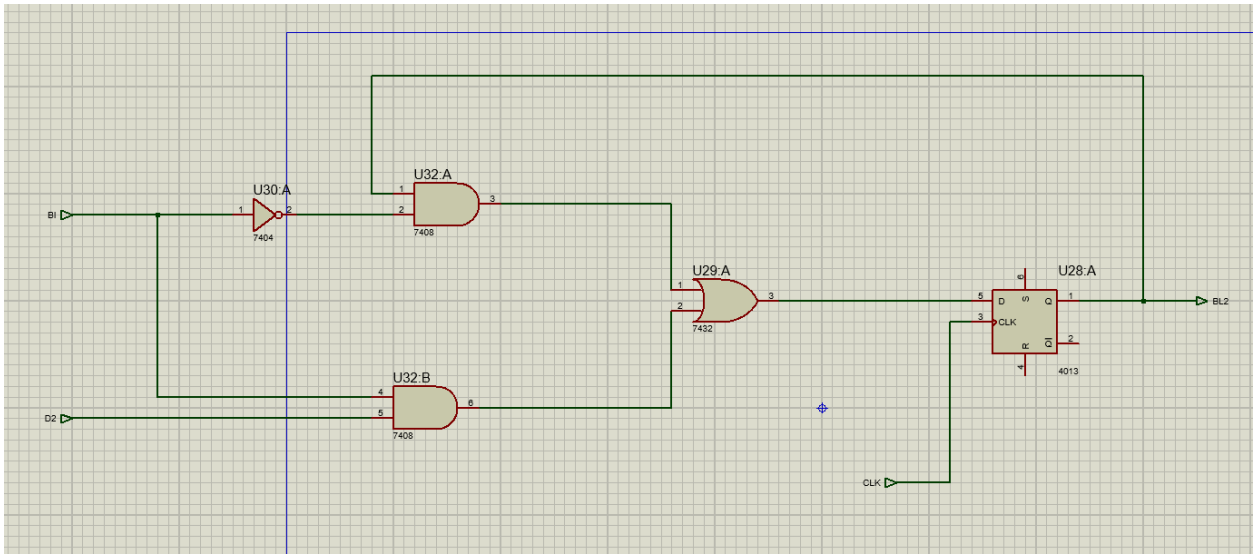
Register B (Implemented using Logic Gates)

Connection in Parent Sheet



In the parent sheet, Register B is also connected to the common bus through D0–D7 for input. It uses the same clock line for synchronization. This register stores data from the bus when its load control is active and holds the value until needed for further operations.

Connection in Child Sheet



Just like the Accumulator, Register B is made of eight 1-bit register cells using logic gates, a 4013 D flip-flop, and a 74126 tri-state buffer. The logic gates control when the input data is allowed into the flip-flop. On the clock edge, the bit is stored. The stored value can then be used internally when required. By combining eight such cells, an 8-bit Register B is created without using any IC.

Brief Description of Circuit Connection Between Accumulator and Register B

The Accumulator and Register B are both built using 8-bit register ICs like the 74LS173. They are connected in parallel to the same 8-bit data bus, which means both receive the same input data from sources like RAM. However, each register has its own separate control signals: Load Enable (E1, E2) and Output Enable (OE1, OE2). This allows the system to independently decide when each register loads data from the bus or outputs data to it.

In the circuit, they work together by passing data between themselves and RAM using the shared bus. For example, data from RAM can be loaded into Register B, then later moved to the enabling the other register's load input, then pulsing the clock. This controlled transfer ensures smooth data flow without bus conflicts, preparing the system for future ALU operations.

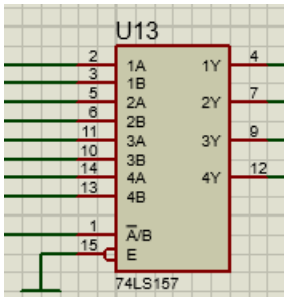
RAM (Random Access Memory)

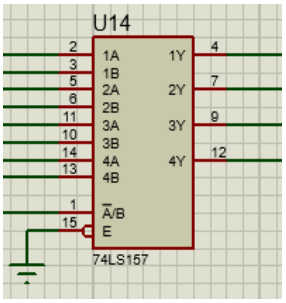
RAM is a 16x8 bit memory device that has a 14-pin interface and is capable of storing 8-bit data words. It features dedicated 8-bit input and 8-bit output buses, four address lines for selecting one of its 16 internal locations, and three control pins: one for selecting read or write mode, and a separate output enable (RO) pin.

IC/Components Used

1. 74LS157

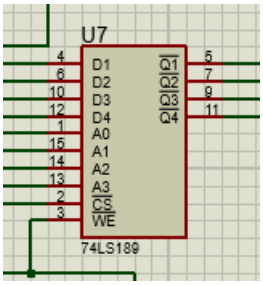
This is a Quad 2-Input Multiplexer. It acts as a switch to decide which set of 4-bit signals gets through to the RAM. It takes two sets of inputs and, based on the Select pin, routes one of those sets to the output. It allows us to toggle between two different data or address sources easily.

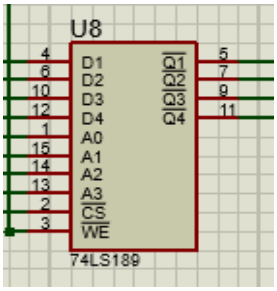




2. 74LS189

This is a 16x4 Static RAM. We have combined them to store 8-bit words. It stores 4 bits of data at 16 different address locations. This chip has Inverted Outputs. If we write a 1, the chip physically holds a 0 at the output pin.

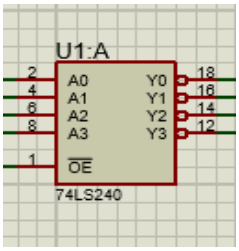


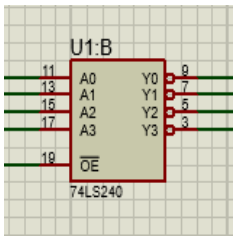


3. 74LS240

This is an Inverting Octal Buffer with Tri-state outputs. It does two things:

1. Inverts the signal: It flips the backward data from the RAM so LEDs show the correct value.
2. Tri-state Control: It can disconnect from the bus entirely, allowing other components to use the lines without interference.



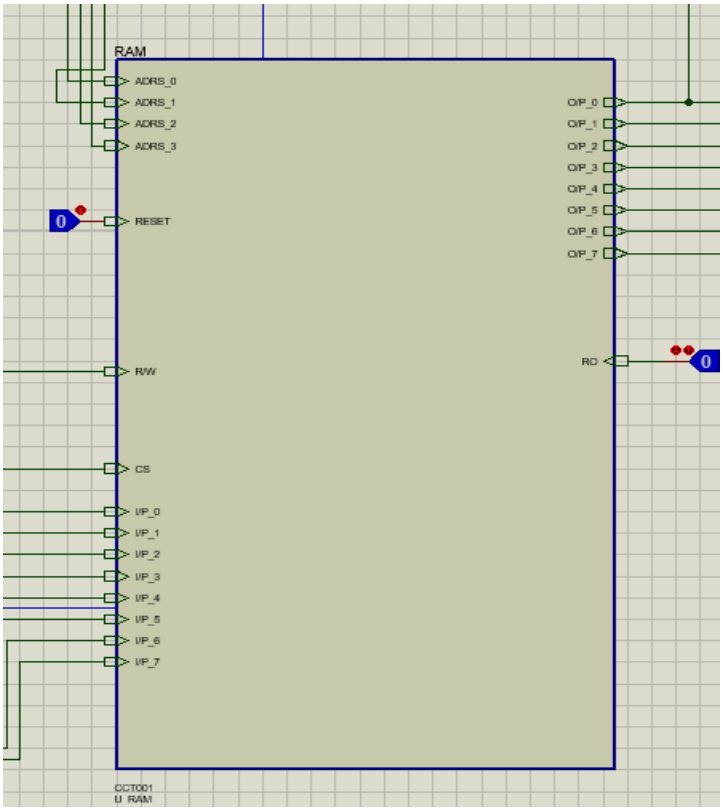


Description of Connections

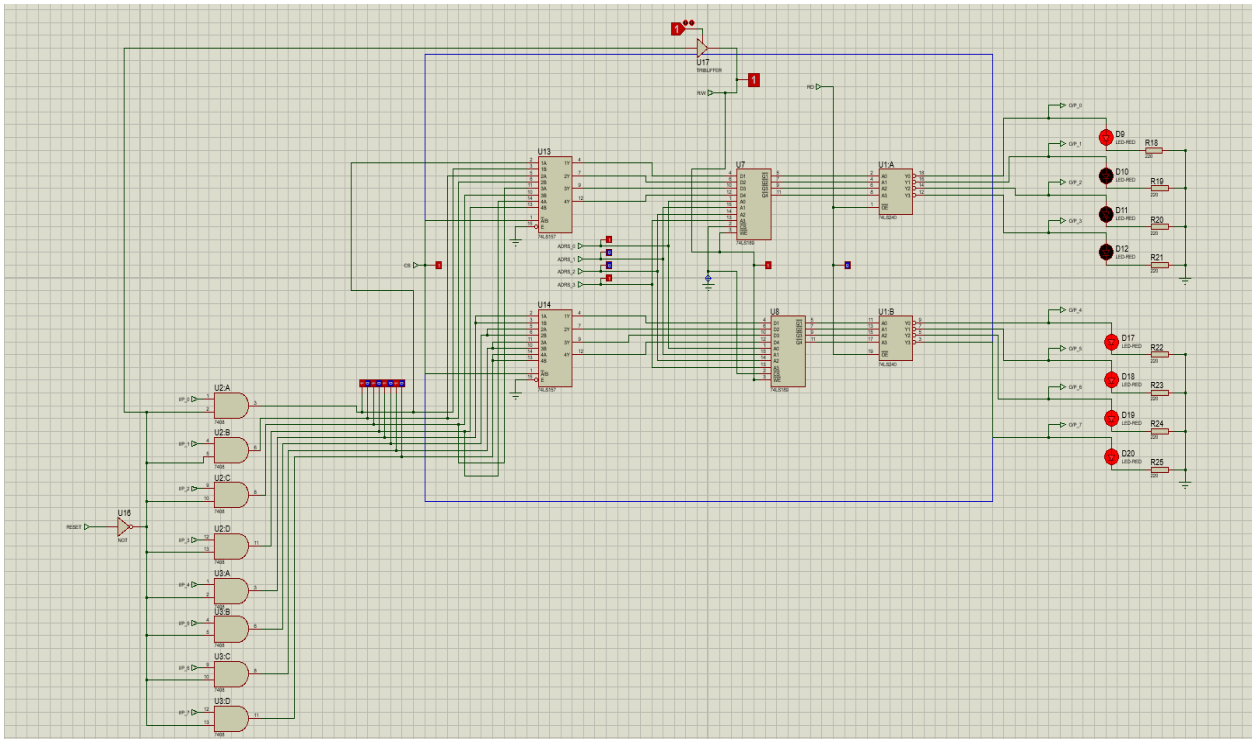
Parent Sheet:

Bus Connections: The parent sheet connects the main System Data Bus (8-bit) and Address Bus (4-bit) to the ports of the child sheet. It acts as the bridge between the user inputs and the internal memory storage.

Control Logic: This level manages the master control signals. These are routed into the child sheet to synchronize the memory's operation with other components.



Child Sheet:

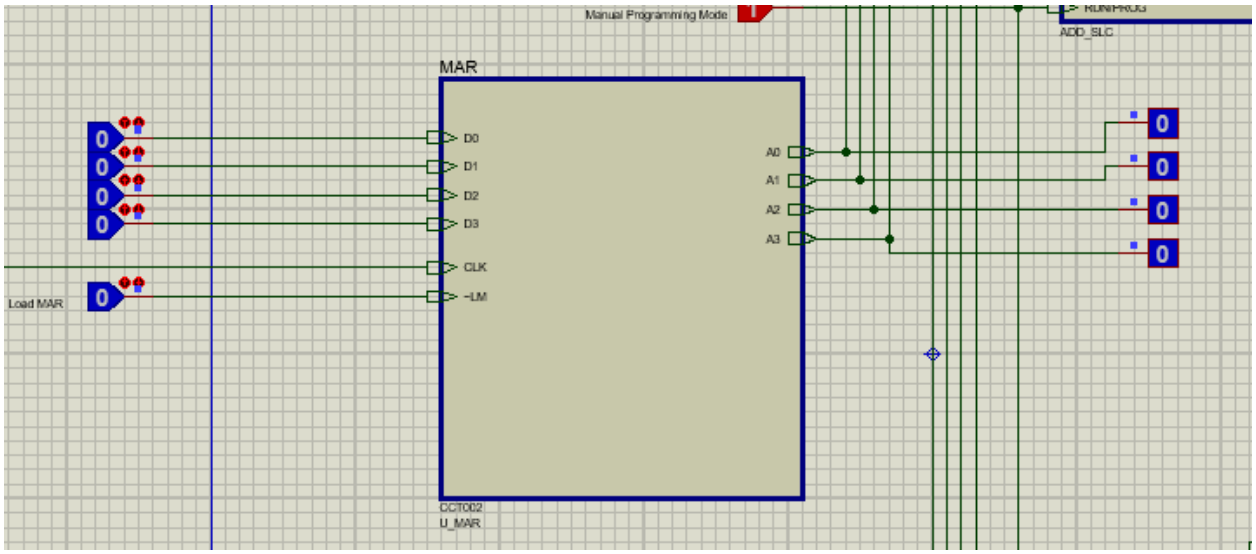


Address & Selection: The **74LS157** multiplexers receive external address signals and route them to the **74LS189** RAM chips. The RAM chips are enabled via the Chip Select (CS) and Write Enable (R/W) lines to determine if data is being stored or accessed.

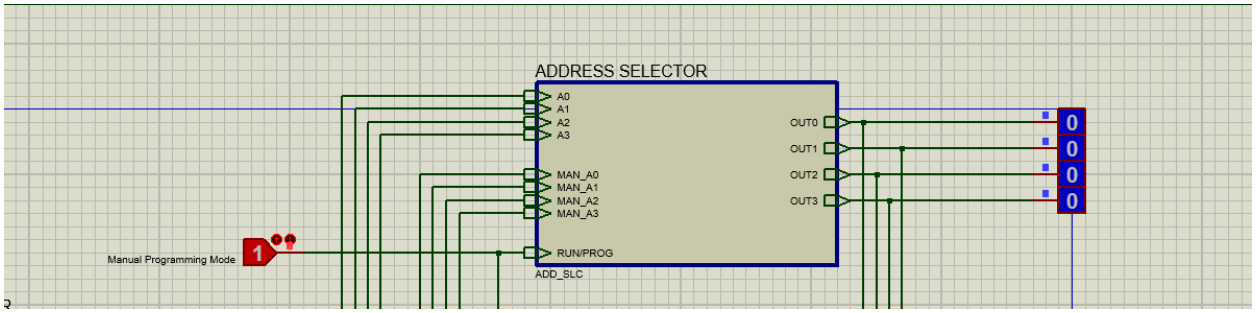
Data Path: Data flows into the RAM chips from the input bus. Because the 74LS189 has **inverted outputs**, the data is then passed through the **74LS240** inverting buffers. This flips the data back to its original state before it reaches the output.

Output Control: The 74LS240 buffers also manage the **Tri-state** condition via the RD (Read) signal. This ensures that the memory module only drives the output LEDs (D9–D20) when specifically instructed to read, preventing bus conflicts.

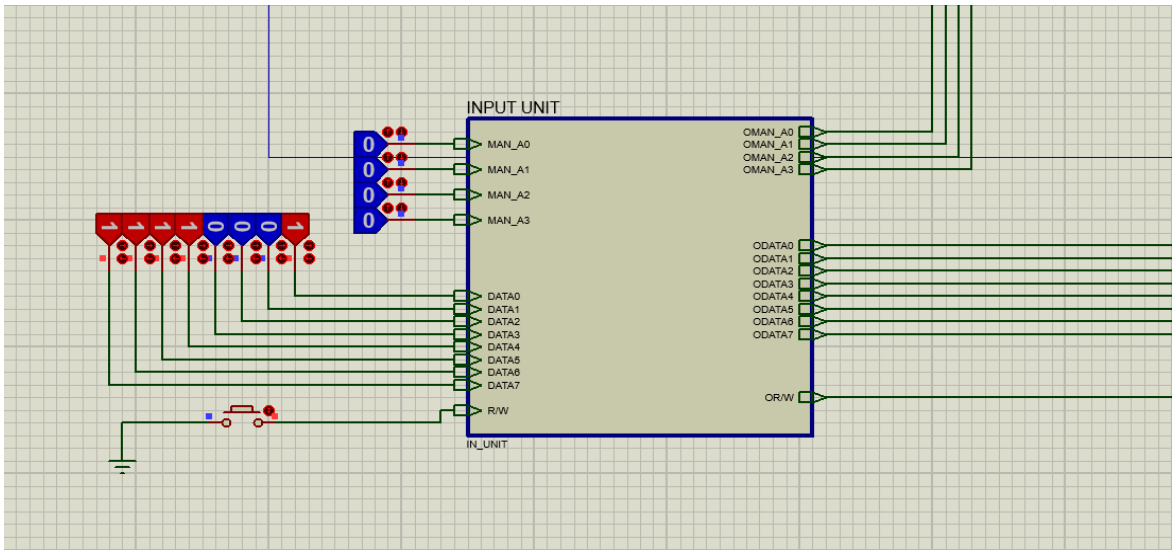
MAR:



Address Selector:



Input Unit:



Functions

Parent Sheet

Address Inputs:

These four lines determine which of the 16 memory locations is being accessed for reading or writing.

Control Lines:

- R/W (Read/Write) line triggers data storage.
- CS (Chip Select) activates the entire block. When CS is ON, the input of the MAR is saved as address memory and when CS is off, the input of the input unit is saved as address memory.
- RO (Read Output) enables the tri-state buffers to send data to the bus.

Data I/O:

These represent the 8-bit input path for storing data and the 8-bit output path for displaying it via the LEDs.

Child Sheet

Multiplexer Routing:

The 74LS157 pins are wired to select between two sets of 4-bit inputs; the outputs feed directly into the Data inputs of the RAM chips.

RAM Storage:

The 74LS189 chips share the Address and Control lines in parallel; U7 handles the lower nibble (0-3) while U8 handles the upper nibble (4-7) to form a complete byte.

Buffer Correction:

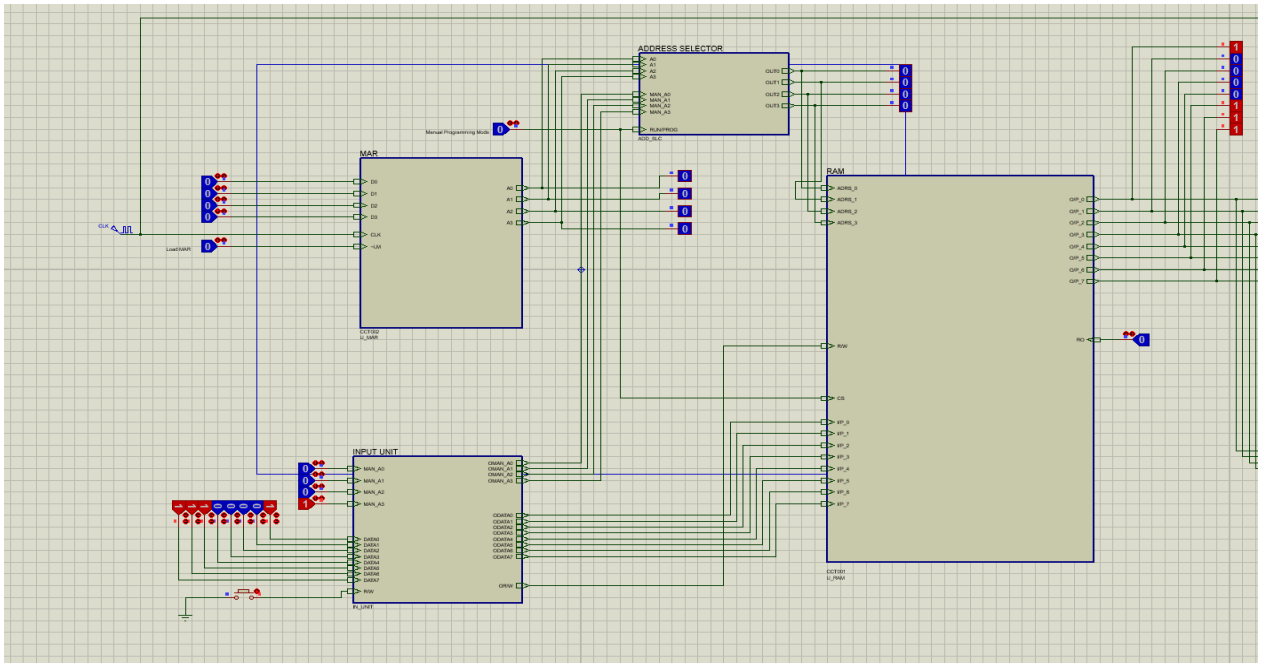
The RAM output pins connect to the 74LS240 inputs; this restores the signal from inverted back to normal before the data reaches the output terminals.

Working Principle

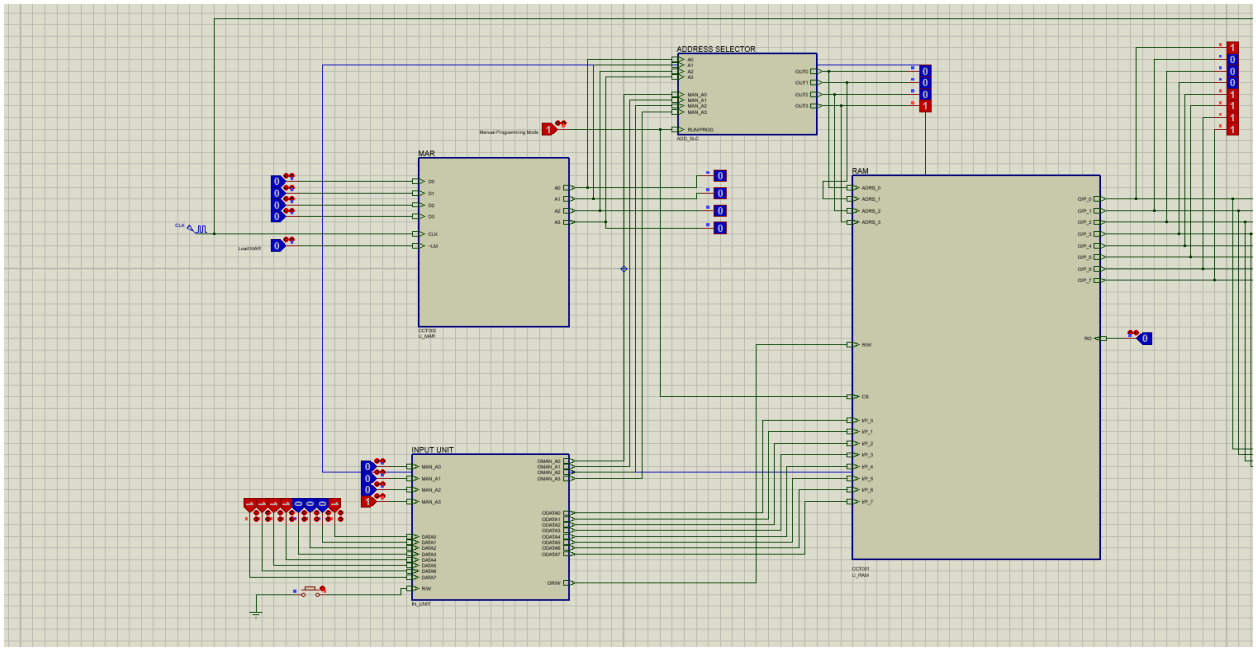
At the Parent Sheet level, the circuit acts as the system's control center, where the Address Selector block determines whether the RAM is controlled by the Memory Address Register (MAR) for automated tasks or the Input Unit for manual programming.

When the Manual Programming Mode is off (0), the Address Selector (multiplexer) is set to route the 4-bit address from the MAR (Memory Address Register) to the RAM. In this mode, if the MAR provides address 0000, the RAM targets that specific memory slot. When the push button (Write Trigger) is pressed, the 8-bit data 10000111 from the Input Unit is written into that address. The RAM captures this

data through the 74LS189 chips, which are enabled by the CS and R/W signals.



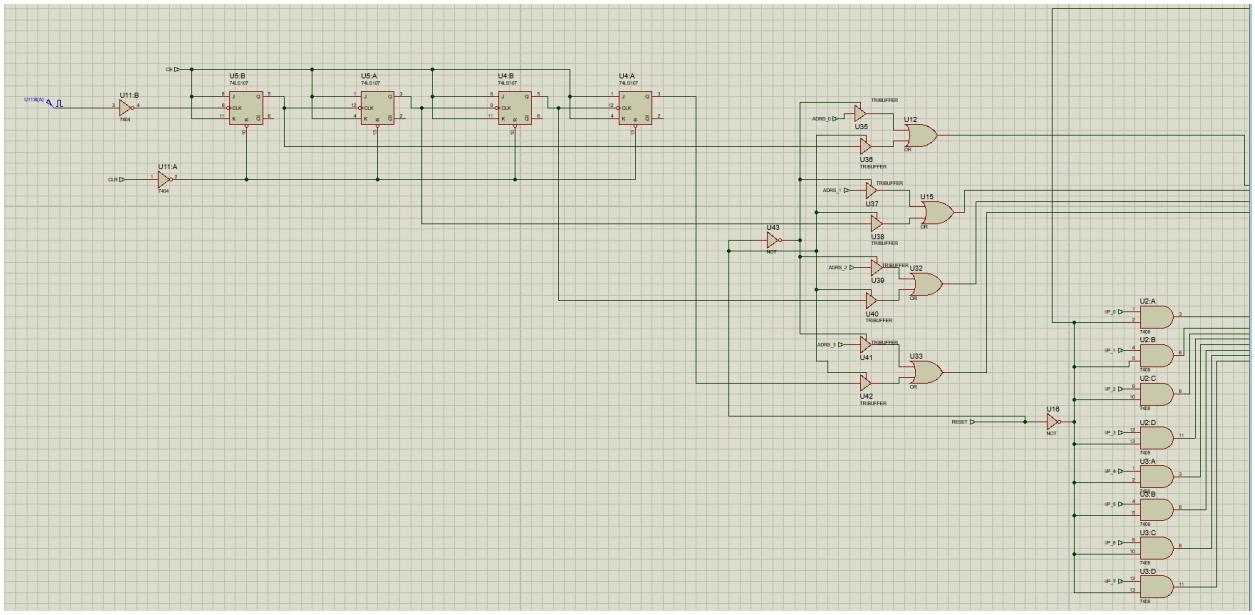
If the Manual Programming Mode is switched on (1), the MAR is bypassed, and the Address Selector now pulls the 4-bit address directly from the Input Unit's manual address toggles. If you set the manual address to 0001 and press the push button, the new 8-bit data string 10001111 is stored at that second memory location. Internally, the 74LS157 multiplexers on the child sheet handle this high-speed switching between the two potential address/data sources to ensure the RAM receives the correct signals without interference.



Once a value is latched into the 74LS189 memory cells, it remains there as long as the circuit has power. When you return to a previous address (like 0000), the RAM outputs that stored value regardless of what is currently on the input data lines, provided the R/W line is high (Read mode). The 74LS240 buffers then flip the RAM's internal inverted data and drive it to the LEDs, allowing us to see the original 10000111 or 10001111 exactly as it was first saved.

Additional Feature:

It is seen that when we need to change all the stored memory values, we need to erase all the data one by one. It is hectic for the user to do that manually. So we tried to build a system where you can make all the saved data to zero by implementing a single logic to the system. With this system you don't need to pause the simulation. You can change all the values to zero while it is running.



The inclusion of a Reset Button significantly improves the simulation workflow by eliminating the need to restart the entire Proteus session to clear memory. When this button is pressed, it triggers a logic sequence, likely through the AND gate array shown in the schematic that overrides the current data lines and forces the 8-bit memory input to a null state. This effectively clears the previously saved data in the 74LS189 RAM chips, resetting the output to 00000000 and allowing us to begin a fresh manual or automated programming cycle immediately.

Combination of RAM and Register:

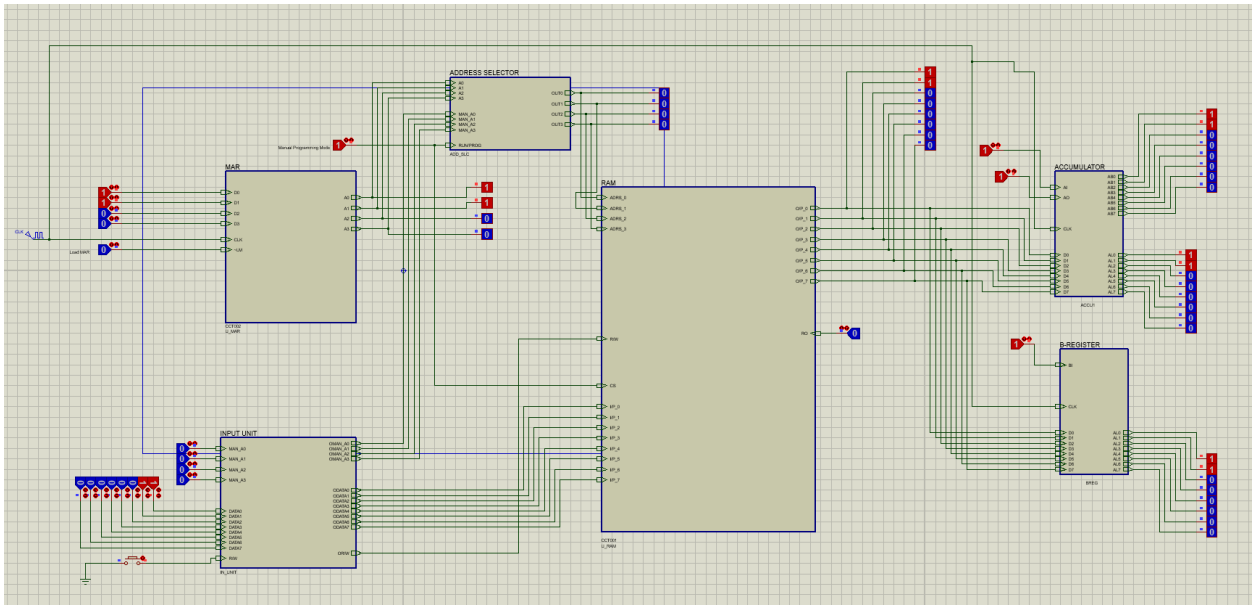
The circuit operates using a single 8-bit common data bus that connects the RAM, Accumulator (Register A), and Register B. This bus allows data to be transferred between components, while

control signals and tri-state buffers ensure that only one module places data on the bus at a time, preventing conflicts.

The RAM stores 8-bit data at selected memory addresses. During a read operation, the data stored at the chosen address is placed onto the system bus. This bus data can then be loaded into either the Accumulator or Register B on the next clock pulse, depending on which register's load control signal is active.

The Accumulator stores data from the bus and can also send its stored value back onto the bus when enabled. This allows it to act as the main working register, holding data from memory and sharing results with other parts of the system. In contrast, Register B only stores data from the bus and sends its output directly to the ALU, without using the shared bus.

Together, RAM supplies data, the bus transfers it, the Accumulator manages and shares working values, and Register B provides a stable second operand for computation. All operations are synchronized by the clock and controlled by enable signals, allowing the circuit to function as a complete SAP-1 memory and register subsystem.



Discussion:

The combined RAM and register circuit successfully demonstrate controlled data storage and transfer using a single shared bus. By carefully managing clock and control signals, the system ensures that data moves between memory and registers without bus conflicts. However, the work became easy after using the 74LS189 IC. Otherwise we had to make 128 memory cells one by one using D flipflops which is time-consuming and less efficient in terms of making SAP-01.

The Accumulator plays a central role by both holding data and placing it back onto the bus when required, while Register B supports computation by providing a dedicated operand to the ALU. This separation of roles keeps the design simple and efficient.

Overall, the circuit follows the SAP-1 architecture closely and provides a solid foundation for integrating the ALU and control unit in later stages of the project.

Credits:

Work Contribution	Contributors
RAM, MAR and Address Selector	Sanajid, Rafin and Raad
B-Register and Accumulator	Rafi
Report Writing	Samiha and Nazifa